

# A Geometric and Bayesian Formulation of Autoregressive Language Models

Akshay Mishra

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## Abstract

This paper formalizes the mechanics of autoregressive Large Language Models (LLMs) by reframing their forward pass and pre-training objectives through the lens of Bayesian inference, high-dimensional geometry, and statistical mechanics. We show that the conditional probability distribution generated by the Transformer architecture can be explicitly decomposed into an intrinsic token prior and a contextual likelihood. We map these probabilistic components directly onto the network’s structural blocks—specifically Multi-Head Attention (MHA) and Feed-Forward Networks (FFN)—and demonstrate how the final output token distribution is modulated by the statistical variance of the latent residual stream, acting as a dynamic inverse temperature. Finally, we formalize pre-training as a self-organizing geometric optimization that resolves these components into a stable semantic coordinate system.

## 1 Introduction and the Bayesian Framing

An autoregressive large language model operates over a finite vocabulary  $V$  [2]. Given a context sequence of  $n$  tokens,  $C = (t_1, t_2, \dots, t_n)$ , where  $t_i \in V$ , the model defines a discrete conditional probability distribution over the vocabulary for the next token  $w \in V$ :

$$P(w \mid C) \tag{1}$$

While modern deep learning architectures evaluate this distribution directly using discriminative causal language modeling [3], we can invert this formulation using Bayes’ theorem to expose the underlying generative dynamics [9]:

$$P(w \mid C) = \frac{P(C \mid w) \cdot P(w)}{P(C)} \tag{2}$$

Because the denominator  $P(C) = \sum_{v \in V} P(C \mid v)P(v)$  represents the marginal likelihood of the context sequence, it acts as a constant normalization factor across all candidate tokens at step  $n + 1$ . Thus, the relative probability mass assigned to any token  $w$  is determined strictly by the numerator:

$$P(w \mid C) \propto P(C \mid w) \cdot P(w) \tag{3}$$

Where:

- $P(w)$  is the **unconditional prior probability** of the token  $w$  occurring in the language, independent of local context.
- $P(C \mid w)$  is the **contextual likelihood**, measuring how strongly the historical sequence  $C$  retroactively aligns with or necessitates the trailing token  $w$ .

## 2 Mathematical Decomposition of Likelihood $P(C \mid w)$

In a Transformer architecture [1], the context sequence  $C$  is transformed into a sequence of continuous vectors. Let  $X \in \mathbb{R}^{n \times d}$  represent the input embedding matrix for the context, where  $d$  is the model dimension. The final hidden state at the terminal position  $n$  after  $L$  layers is denoted as  $h_n^{(L)} \in \mathbb{R}^d$ .

The unnormalized log-probabilities (logits)  $z_w$  for each token  $w$  are computed by projecting  $h_n^{(L)}$  onto an unembedding matrix  $U \in \mathbb{R}^{|V| \times d}$ , such that  $z_w = h_n^{(L)} \cdot u_w$ , where  $u_w \in \mathbb{R}^d$  is the target token’s unembedding vector [4].

The log-likelihood  $\log P(C \mid w)$  is shaped by the interaction of two foundational architectural components within each layer  $l$ : Multi-Head Attention and the position-wise Feed-Forward Network.

### 2.1 Multi-Head Attention as Contextual Routing

The Multi-Head Attention block at layer  $l$  and position  $n$  computes a dynamically weighted combination of all preceding token states [1]. Let  $H$  be the number of attention heads, and  $d_k$  be the head dimension. For a single attention head  $g \in \{1, \dots, H\}$ , let  $W_Q^{(l,g)}, W_K^{(l,g)}, W_V^{(l,g)} \in \mathbb{R}^{d \times d_k}$  be the Query, Key, and Value projection matrices, respectively.

The terminal query vector for head  $g$  is defined as  $q_n^{(l,g)} = h_n^{(l-1)} W_Q^{(l,g)}$ , while the keys and values for all historical positions  $i \in \{1, \dots, n\}$  are  $k_i^{(l,g)} = h_i^{(l-1)} W_K^{(l,g)}$  and  $v_i^{(l,g)} = h_i^{(l-1)} W_V^{(l,g)}$ . The dynamic attention weights  $\alpha_{n,i}^{(l,g)}$  formalize the conditional token-to-token dependencies:

$$\alpha_{n,i}^{(l,g)} = \frac{\exp\left(\frac{q_n^{(l,g)} \cdot k_i^{(l,g)}}{\sqrt{d_k}}\right)}{\sum_{j=1}^n \exp\left(\frac{q_n^{(l,g)} \cdot k_j^{(l,g)}}{\sqrt{d_k}}\right)} \quad (4)$$

The concatenated outputs of all heads are projected back into the residual stream via the global output matrix  $W_O^{(l)} \in \mathbb{R}^{d \times d}$ . By partitioning  $W_O^{(l)} = [(W_{O,1}^{(l)})^T \dots (W_{O,H}^{(l)})^T]^T$ , where each head’s output projection slice is  $W_{O,g}^{(l)} \in \mathbb{R}^{d_k \times d}$ , the total attention contribution to the log-likelihood of a target token  $w$  decomposes into a sum of bilinear forms:

$$\log P(C \mid w)_{\text{MHA}} \sim \sum_{g=1}^H \sum_{i=1}^n \alpha_{n,i}^{(l,g)} \left( h_i^{(l-1)} W_V^{(l,g)} W_{O,g}^{(l)} u_w \right) \quad (5)$$

This isolates two distinct mechanisms:  $\alpha_{n,i}^{(l,g)}$  handles the contextual routing (determining which past tokens matter), while the dot product acts as a semantic alignment function between the historical features of token  $i$  and the target token  $w$ .

### 2.2 Feed-Forward Networks as Concept Memory Retrieval

Following the attention block, the updated hidden state  $x_n^{(l)} = h_n^{(l-1)} + a_n^{(l)}$  enters the position-wise Feed-Forward Network (FFN). The FFN operates as a key-value associative memory [5]. Let  $W_1^{(l)} \in \mathbb{R}^{d \times d_{ff}}$  represent a bank of intermediate keys  $\{k_m^{(l)}\}_{m=1}^{d_{ff}}$ , and  $W_2^{(l)} \in \mathbb{R}^{d_{ff} \times d}$  represent a bank of intermediate values  $\{v_m^{(l)}\}_{m=1}^{d_{ff}}$ .

Applying a non-linear activation function  $\sigma$  (e.g., GELU [6]), the operation at the terminal position is:

$$\text{FFN}(x_n^{(l)}) = \sum_{m=1}^{d_{ff}} \sigma\left(x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)}\right) v_m^{(l)} \quad (6)$$

Projecting this operation onto the target token embedding  $u_w$  reveals the FFN’s mathematical formulation of the likelihood:

$$\log P(C \mid w)_{\text{FFN}} \sim \sum_{m=1}^{d_{\text{ff}}} \sigma \left( x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right) \cdot (v_m^{(l)} \cdot u_w) \quad (7)$$

Here,  $\sigma \left( x_n^{(l)} \cdot k_m^{(l)} + b_{1,m}^{(l)} \right)$  scales how strongly the synthesized context satisfies the  $m$ -th latent conceptual rule, while  $(v_m^{(l)} \cdot u_w)$  dictates the historical co-occurrence correlation between that rule and the literal vocabulary token  $w$ .

### 3 Geometric Formulation of the Prior $P(w)$ and Logit Mechanics

The exact computation of the unconditional prior  $P(w)$  requires marginalizing over the infinite space of all possible context sequences  $\mathcal{C}$ :

$$P(w) = \sum_{C \in \mathcal{C}} P(w \mid C) P(C) \quad (8)$$

Transformers approximate this expectation statically within their parameters via two geometric mechanics:

#### 3.1 Unembedding Bias ( $b_U$ )

In architectures featuring an explicit unembedding bias  $b_U \in \mathbb{R}^{|V|}$ , the logit calculation is defined as  $z_w = (h_n^{(L)} \cdot u_w) + b_{U,w}$ . Under an uninformative context where  $h_n \rightarrow \mathbf{0}$ , the logit collapses directly to the scalar bias:

$$P(w) \approx \frac{\exp(b_{U,w})}{\sum_{v \in V} \exp(b_{U,v})} \quad (9)$$

#### 3.2 Embedding Norms ( $\|u_w\|$ )

In bias-free architectures, the prior is encoded in the Euclidean magnitude of the token vector,  $\|u_w\|$ . Decomposing the final dot product geometrically yields:

$$z_w = \|h_n^{(L)}\| \|u_w\| \cos(\theta_w) \quad (10)$$

Where  $\theta_w$  is the high-dimensional angle between the final context state and the token vector. Tokens with high baseline frequencies in the training corpus develop longer vector magnitudes  $\|u_w\|$  during optimization, amplifying their baseline logit regardless of the contextual direction  $\cos(\theta_w)$  [7].

## 4 Statistical Mechanics of the Latent Stream

The remaining term in the geometric logit decomposition is the norm of the final hidden state,  $\|h_n^{(L)}\|$ . This component functions as a dynamic, context-dependent entropy controller.

#### 4.1 Activation Variance as Inverse Temperature

Let  $h_n = [x_1, x_2, \dots, x_d] \in \mathbb{R}^d$  be a row vector representing the final hidden state  $h_n^{(L)}$ . If we treat the individual channel activations as samples from an internal distribution with a mean driven to zero by layer normalization ( $\mu_x \approx 0$ ) [8], the cross-channel variance is:

$$\sigma^2 = \frac{1}{d} \sum_{i=1}^d (x_i - \mu_x)^2 \approx \frac{1}{d} \sum_{i=1}^d x_i^2 \quad (11)$$

Because the squared Euclidean norm is  $\|h_n\|^2 = \sum_{i=1}^d x_i^2$ , the vector magnitude can be expressed as a function of activation variance:

$$\|h_n\| = \sqrt{d \cdot \sigma^2} \quad (12)$$

When substituting this back into the vocabulary-wide Softmax function,  $\|h_n\|$  maps perfectly to the **inverse temperature** parameter  $\beta = \frac{1}{T}$  from statistical mechanics [9, 10]:

$$P(w \mid C) = \frac{\exp(\beta \cdot \|u_w\| \cos(\theta_w))}{\sum_{v \in V} \exp(\beta \cdot \|u_v\| \cos(\theta_v))}, \quad \text{where } \beta = \|h_n\| \quad (13)$$

- **High Contextual Certainty:** When the context contains a highly predictable pattern, successive attention heads and FFN layers write aligned information into the residual stream. This constructive interference inflates the cross-channel variance  $\sigma^2$ , causing  $\|h_n\|$  to expand. This yields a high  $\beta$ , sharpening the distribution and lowering its entropy.
- **High Contextual Ambiguity:** When the context is entropic or transitions between distinct topics, architectural sub-layers write orthogonal or destructive vectors into the stream. The variance  $\sigma^2$  collapses, shrinking  $\|h_n\|$ . This yields a low  $\beta$ , flattening the distribution into a high-entropy state.

#### 4.2 Cosine Similarity as a Correlation Coefficient

Similarly, treating the dimensions of  $h_n$  and  $u_w$  as zero-mean random variables allows us to map the likelihood component  $\cos(\theta_w)$  directly to the **Pearson correlation coefficient** ( $r$ ) in feature space:

$$\cos(\theta_w) = \frac{\text{Cov}(h_n, u_w)}{\sigma_{h_n} \sigma_{u_w}} = r_{h_n, u_w} \quad (14)$$

where  $\sigma_{h_n}$  and  $\sigma_{u_w}$  denote the standard deviations of the components of  $h_n$  and  $u_w$ , respectively. Thus, the unnormalized logit is exactly proportional to the total feature co-variance scaled by the model dimension:  $z_w = \text{Cov}(h_n, u_w) \cdot d$ .

### 5 Statistical Description of Pre-Training

During inference,  $u_w$  is static. During pre-training, both the context state  $h_n$  and the vocabulary matrix  $U$  are discovered simultaneously via the minimization of the cross-entropy loss [11] for a target next-token  $w^*$ :

$$\mathcal{L} = -\log \frac{\exp(h_n \cdot u_{w^*})}{\sum_{v \in V} \exp(h_n \cdot u_v)} \quad (15)$$

This framework can be understood as a dual-force geometric optimization process:

### 5.1 The Mutual Attraction Force

The gradient with respect to the correct token vector  $u_{w^*}$  acts as an attractive force:

$$\frac{\partial \mathcal{L}}{\partial u_{w^*}} = -(1 - P(w^* | C))h_n \quad (16)$$

This force dynamically rotates  $u_{w^*}$  and the context representation  $h_n$  toward one another, maximizing their directional correlation  $\cos(\theta_{w^*})$ . Simultaneously, if a token  $w^*$  is consistently validated across diverse contexts, its vector norm  $\|u_{w^*}\|$  is driven upward to minimize loss across the corpus, mapping out its global prior  $P(w)$ .

### 5.2 The Global Repulsion Force

Conversely, the denominator of the Softmax generates a contrastive repulsion force against all incorrect tokens  $v \neq w^*$ , resembling contrastive representation learning frameworks [12]:

$$\frac{\partial \mathcal{L}}{\partial u_v} = P(v | C)h_n \quad (17)$$

This pushes the context state  $h_n$  away from incorrect vocabulary vectors, forcing them toward orthogonality ( $\cos(\theta_v) \rightarrow 0$ ).

### 5.3 Emergence of Distributional Semantics

Because the network minimizes this loss across billions of tokens, semantic clustering emerges naturally. If two distinct tokens  $w_1$  and  $w_2$  share high contextual substitutability, they will repeatedly receive identical directional updates from shared context states  $h_n$ . This self-organizing system settles into a geometric steady-state where vocabulary vectors discover a coordinate system whose spatial proximity mirrors latent semantic relationships, validating the distributional hypothesis [13]:

$$\cos(\theta_{w_1, w_2}) \rightarrow 1 \iff \mathcal{C}_{w_1} \approx \mathcal{C}_{w_2} \quad (18)$$

where  $\mathcal{C}_w$  represents the distribution of contexts in which token  $w$  appears.

## 6 Conclusion

By formalizing the Large Language Model as a Bayesian engine, we transition from a qualitative understanding of neural network behaviors to an exact geometric framework. Multi-Head Attention and Feed-Forward Networks act in tandem to iteratively construct a context vector whose final magnitude represents the model’s statistical certainty ( $\beta$ ), and whose orientation represents structural feature configurations. The final token prediction is thus a direct measure of the Pearson correlation between this dynamic contextual variance and the self-organized coordinate landscape of the vocabulary embeddings.

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